

Maithili Patel

POSTDOCTORAL RESEARCHER · ROBOTICS · CARNEGIE MELLON UNIVERSITY

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Research Interests

I am a postdoctoral researcher conducting research at the intersection of artificial intelligence and human-robot interaction. I aim to pursue an academic position to continue advancing research in the field, and support others interested in similar topics.

Education

GEORGIA INSTITUTE OF TECHNOLOGY · *PhD, Robotics (Interactive Computing)*

2021 - 2026

- Advised by Prof. Sonia Chernova

UNIVERSITY OF MICHIGAN · *MSE, Mechanical Engineering (Robotics)*

2017 - 2019

- Advised by Prof. Chad Jenkins

INDIAN INSTITUTE OF TECHNOLOGY, BOMBAY · *B.Tech, Mechanical Engineering*

2013 - 2017

- Graduated with Honors, holding a Minor in Computer Science and Engineering

Publications

JOURNAL

- A Survey of Semantic Reasoning Frameworks for Robotic Systems, W. Liu*, A. Daruna*, **M. Patel****, K. Ramachandruni**, S. Chernova
Robotics and Autonomous Systems (RAS), 2022

CONFERENCE

- ADAPT: Actively Discovering and Adapting to Preferences for any Task, **M. Patel**, X. Puig, R. Desai, R. Mottaghi, S. Chernova, J. Truong, A. Rai
Conference on Language Modeling (COLM), 2025
- A Model-Agnostic Approach for Semantically Driven Disambiguation in Human-Robot Interaction, F. Dogan, **M. Patel**, W. Liu, I. Leite, S. Chernova
International Conference on Robot and Human Interactive Communication (IEEE RO-MAN) 2025
- Robot Behavior Personalization from Sparse User Feedback, **M. Patel**, S. Chernova
Robotics and Automation Letters (IEEE RAL) 2025
- Predicting Routine Object Usage for Proactive Robot Assistance, **M. Patel**, A. Prakash, S. Chernova
Conference on Robot Learning (CoRL), 2023
- Proactive Robot Assistance via Spatio Temporal Object Tracking, **M. Patel**, S. Chernova
Conference on Robot Learning (CoRL), 2022

WORKSHOP

- Longitudinal Proactive Robot Assistance,
M. Patel
HRI Pioneers, 2023 workshop
- Understanding In-home Routines through Spatio-temporal Object Tracking for Proactive Assistance,
M. Patel, S. Chernova
ICRA, 2022 workshop - Prediction and Anticipation Reasoning in Human Robot Interaction
- Robot Behavior Personalization from Sparse User Feedback,
M. Patel, S. Chernova
ICRA, 2025 workshop - Human-Centered Robot Learning in the Era of Big Data and Large Models

Academic Achievements and Service

- Fellowship awardee of the the Carnegie Bosch Institute (CBI) Postdoctoral Fellowship program, 2026 2026-28
- Student Volunteer for IEEE International Conference on Robotics and Automation (ICRA), 2025 May 2025
- Area Chair for late breaking reports at the ACM/IEEE International Conference on Human Robot Interaction (HRI), 2025 and 2026 2024-26
- Participant in Doctoral Consortium at the Summit on Responsible Computing, AI, and Society, Atlanta October 2024
- Invited speaker at the Environment Dynamics Matters workshop at IROS, 2024, Abu Dhabi. October 2024
- Reviewer for RSS, CoRL, HRI, RAL, T-HRI, COLM, ICRA, IJHCI, ICSR, UR, HRI-Pioneers, and RSS-Pioneers 2022-Present
- Fellowship awardee of the Georgia Robotics Fellowship program. April 2024
- Student Volunteer Coordinator for Conference on Robot Learning, 2023 November 2023
- HRI Pioneer at the ACM/IEEE International Conference on Human Robot Interaction (HRI), 2023. March 2023
- Co-organizer for the Semantic Scene Understanding for Human Robot Interaction workshop at the ACM/IEEE International Conference on Human Robot Interaction (HRI), 2023. March 2023
- Participant in Google CS Research Mentorship Program to receive mentorship from a Googler and network with other participants to further my career in computing 2022-2023
- Participant in RPL Summer School, 2022, Stockholm, Sweden, fully funded and organized by the Division of Robotics, Perception and Learning at KTH Royal University, Sweden June 2022

Research Experience

META · Research Scientist Intern, FAIR

May 2024 - Aug 2024

Socially Intelligent Robots under Embodied AI at Fundamental AI Research (FAIR) Labs.

- Formulated and developed a method to address personalized adaptation of robot behavior from sparse dialog.

ALLEN INSTITUTE FOR AI (AI2) · *Research Intern, Mosaic*

May 2023 - Aug 2023

Mosaic at AI2, led by Prof. Yejin Choi, conducts research on Commonsense Reasoning in AI systems.

- Formulated the problem of goal inference from visual observations of partially complete tasks.
- Developing a method involving commonsense reasoning with language models and visual grounding models for video understanding.

THE LABORATORY FOR PROGRESS, UNIVERSITY OF MICHIGAN · *Graduate Researcher*

Advisor: Prof. Chad Jenkins

Planning over affordance wayfields

Jan 2019 - Apr 2019

- Explored null-space subsumption architecture to create planners over potential-based affordance representations achieve constrained manipulation tasks. Demonstrated pick and place on a coffee mug without spilling its contents, using OpenRAVE to simulate and visualize.

Potential Field guided RRTs for motion planning

Jan 2019 - Apr 2019

- Created a hybrid motion planner based on RRT and potential field based gradient descent and achieved greater obstacle-clearance and faster convergence compared to a vanilla RRT.

Manipulation over Mobile Robot Platform

Jan 2018 - Apr 2018

- Implemented capability for a Fetch robot to open doors using MoveIt package on ROS platform, and integrated it into a mobile manipulation pipeline based on a behavior tree structure.

Industry Experience

TOYOTA RESEARCH INSTITUTE · *Software Engineer, Automated Mapping Platform*

Jan 2021 - July 2021

At TRI, the research arm of Toyota, I worked on a mapping project with the Autonomous Driving team.

- Made improvements to the algorithms used to probabilistically infer object changes in a SLAM-based map over time.
- Developed a validation toolkit for evaluating metric accuracy of feature-based maps against a reference, and evaluating precision-recall over detected changes between an old and new map.
- Helped the Women and Allies ERG host Women's History Month to spread awareness and support women's progress.

VEONEER · *Algorithm Engineer*

June 2019 - Jan 2021

Veoneer, formerly Zenuity, develops AD and ADAS software. I worked on Motion Planning and Localization teams based in US.

- Tested mapping and localization by simulating worst-case scenarios, and improved lane curvature calculation.
- Implemented code to identify roadside barriers in HD-maps and extracting a polygon representation of free space.
- Initiated and ran a Journal Club for the team to gain and share domain knowledge through literature discussions.

ZENUITY · *Intern, Decision & Control*

May 2018 - Aug 2018

- Improved path-tracking performance of the lateral dynamics controller in simulation and on vehicle.
- Identified challenges of autonomous parking in an indoor parking structure and created prototypes of decision trees for high-level behaviour planning and kinodynamic A* based motion planning.

Teaching And Mentorship

ROBOT AUTONOMY AND INTERACTIVE LEARNING (RAIL) LAB · Mentor

Fall 2022 - Present

- Helped mentor a team of 2 graduate students towards building a semantic mapping software stack on a Hello Robot Stretch manipulator.
- Helped mentor a team of 4 graduate students towards building a mobile manipulation software stack on a Hello Robot Stretch manipulator.
- Mentored two graduate students through independent research in semantic robotics towards in-home assistance.

IC BUDDIES PROGRAM · Mentor

Fall 2025

- Providing mentorship to 3 junior PhD students towards managing work,

GEORGIA INSTITUTE OF TECHNOLOGY, ATLANTA · Teaching Assistant

- **Human Robot Interaction** with Prof. Sonia Chernova Fall 2024 and 2025
Graduate course on human robot interaction, based on seminar-style paper discussions and final project
- **Introduction to Autonomous Robotics** with Prof. Seth Hutchinson and Prof. Sonia Chernova Spring 2024
Introductory graduate course on autonomous robotics including perception, localization, and planning.

HELEN SCHOLARS PROGRAM (NCWIT) · Mentor

Fall 2021-Spring 2022

- Mentoring an undergraduate student towards professional and personal development as a part of a pilot scholarship program by the National Center for Women & Information Technology (NCWIT)

UNIVERSITY OF MICHIGAN, ANN ARBOR · Graduate Student Instructor

Received honorable mention for **2020 Richard and Eleanor Towner Prize** for Outstanding GSIs

- **Mobile Robotics** with Prof. Maani Jadidi Spring 2019
Graduate-level course focusing on probabilistic methods in robotics like Bayesian methods and Graphical formulations, for perception and SLAM
- **Introduction to Autonomous Robotics** with Prof. Chad Jenkins Fall 2018
Graduate-level course on Autonomous Robotics focusing on planning, covering path planning, manipulation and controls

INDIAN INSTITUTE OF TECHNOLOGY, BOMBAY · Teaching Assistant

- Biology : Qualitative and quantitative models in life systems and applications of engineering in biology Fall 2014